

SMART INDIA HACKATHON 2026



TITLE PAGE

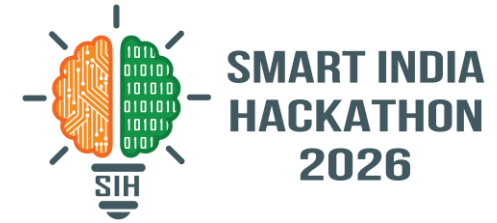
- **Problem Statement ID** – 26142
- **Problem Statement Title** – Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries
- **Theme** – Space Technology
- **PS Category** – Software
- **Team ID** – 190286
- **Team Name** – Sentinels_W



Live prototype
scan to open

PROBLEM & SOLUTION OVERVIEW

Sentinels_W



PS 26142 · NTRO · SPACE TECHNOLOGY

Deep-learning super-resolution mapping from medium-resolution satellite imagery

Asks for: Sentinel-2 10 m $\rightarrow$ < 4 m · geo & spectral consistency · explicit uncertainty · validation against HR reference

PROBLEM

10 m pixels hide small buildings, narrow roads, field bunds and localised damage, which limits classification, change detection and response.

GAP

Commercial VHR is costly and must be tasked. Generic SR hallucinates, alters radiometry and reports no confidence.

SOLUTION

TRINETRA-SR: physics-constrained $4\times$ SR on all 10 bands, a per-pixel trust map, and it runs anywhere, even offline.

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- **Input:** Sentinel-2 L2A, 10 bands (B2–B8A, B11, B12), chosen by cloud-aware search or uploaded as a GeoTIFF.
- **Model:** deep SR backbone + **low-frequency hard constraint** (FFT), so the network can only add detail, never change measured radiometry.
- **Output:** 2.5 m 10-band GeoTIFF + σ -uncertainty GeoTIFF + validation metrics, in the source UTM grid.

How it addresses the problem

- **Fine-scale interpretation:** roofs, stadiums, bunds and debris tracks resolved ($1.17\times$ edge energy vs bicubic).
- **Consistency by construction:** degrading the output reproduces the input at **51.1 dB (10 m) / 36.8 dB (20 m)**, spectral angle 0.48 degrees.
- **Uncertainty handled:** inferred detail is flagged pixel by pixel; mean $\sigma = 0.0008$ reflectance.

❖ Innovation and uniqueness of the solution



Trust map

σ layer + validation on every product



Physics-consistent

FFT hard constraint keeps radiometry



All-band 2.5 m

NDVI · NDRE · NBR at 2.5 m



Zero-server

19 MB, runs offline / air-gapped



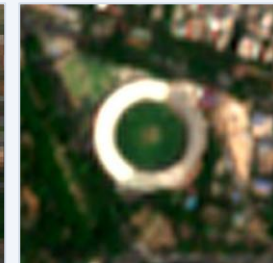
Multi-temporal + conformal

observed detail, guaranteed intervals (v2)

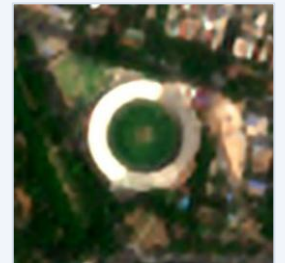
Prototype output · M. Chinnaswamy Stadium, Bengaluru · 13 Mar 2024



Sentinel-2 · 10 m



Bicubic · 2.5 m



TRINETRA-SR · 2.5 m

4×

resolution

10

bands

51 dB

consistency

19 MB

model

Tech stack

Python

PyTorch

ONNX Runtime Web

WASM / WebGPU

Rasterio · GDAL

STAC · COG

Three.js

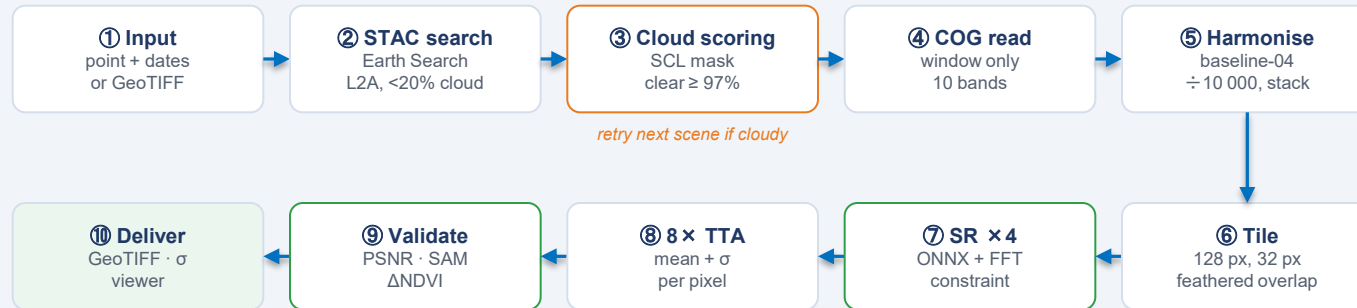
Leaflet

FastAPI · Docker

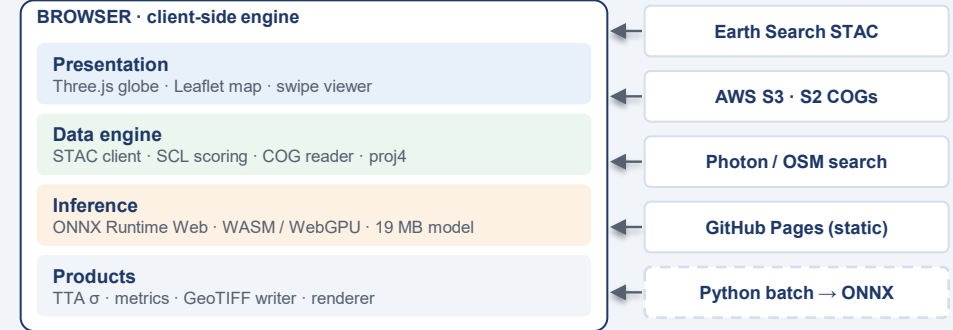
GitHub Pages

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)

End-to-end flowchart



System architecture



❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

Tech stack

AI Python · PyTorch · SEN2SR-Lite · ONNX
Geo Rasterio · GDAL · STAC · COG · proj4
Web JavaScript · ONNX Runtime Web · Three.js · Leaflet · geotiff.js
Ops GitHub Pages · FastAPI · Docker
HW Any laptop CPU; GPU optional (training / batch)

Input / Output

INPUT
 S2 L2A · 10 bands
 4 × 10 m + 6 × 20 m
 AOI point + dates, or
 GeoTIFF (10/12/13 bands)

OUTPUT
 2.5 m GeoTIFF · 10 bands
 σ-uncertainty GeoTIFF
 metrics JSON · viewer
 source UTM, reflectance × 10⁴

Deployment

Web app (client-side)
 GitHub Pages; nothing uploaded **BUILT**

Offline / air-gapped
 serve docs/ + internal COG mirror **BUILT**

On-prem API
 FastAPI + Docker **BUILT**

Batch production
 Python tiler, GPU optional **BUILT**

NTRO integration
 STAC mirror, QGIS plug-in **V2**

Features

- ✓ Search any place · use my location
- ✓ Cloud-aware scene selection
- ✓ Swipe viewer · zoom · full screen
- ✓ NDVI 10 m vs 2.5 m
- ✓ Uncertainty overlay
- ✓ Auto validation metrics
- ✓ GeoTIFF export (3 layers)
- ✓ Upload own GeoTIFF
- ✓ 3-D globe · 5 study areas

Links

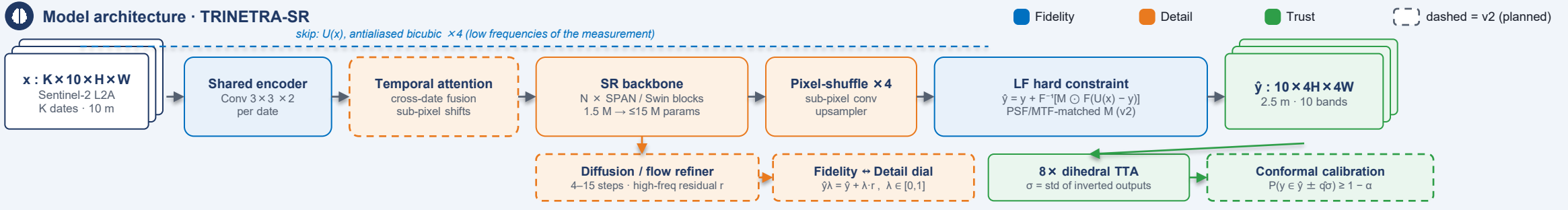
Web app
 scan or open the link below

Web app
uchihaDZoro.github.io/Deep-Learning-Super-Resolution-Mapping

GitHub
github.com/UchihaDZoro/Deep-Learning-Super-Resolution-Mapping

Technical design
[TECHNICAL_DESIGN.md](#)

Demo video
[\[add YouTube link\]](#)

**1 · FIDELITY**

measured information is never altered

4 × super-resolution $10 \text{ m} \rightarrow 2.5 \text{ m}$ ground sampling, $16 \times$ pixels

BUILT

Low-frequency constraint

FFT hard layer restores the input's low band

BUILT

PSF / MTF consistency

per-band Gaussian PSF from the S2 MTF spec

V2

Spectral consistencySAM 0.48° (10 m) · ΔNDVI 0.007 measured

BUILT

All-band reconstruction

B2–B8A, B11, B12 to 2.5 m, SWIR via fusion

BUILT

2 · DETAIL

recover structure a 10 m pixel hides

Deep SR backboneCNN + SPAN attention (v1); Swin / Mamba $\leq 15 \text{ M}$ (v2)

BUILT

Temporal attention

fuse 4–8 revisits; sub-pixel shifts add real detail

V2

High-frequency reconstructionresidual learning over $U(x)$; $1.17 \times$ edge energy

BUILT

Diffusion / flow refinement

optional residual refiner, 4–15 steps

V2

Fidelity ↔ Detail controluser dial λ blends fidelity and refiner outputs

V2

3 · TRUST

every pixel says how sure it is

Uncertainty estimationper-band σ layer shipped as GeoTIFF

BUILT

TTA / ensemble8 rotations and flips; $\sigma =$ spread of outputs

BUILT

Conformal calibrationsplit-conformal $q^* \rightarrow$ intervals with $\geq 1 - \alpha$ coverage

V2

Hallucination detectionphantom-object audit vs HR; must fall in high σ

V2

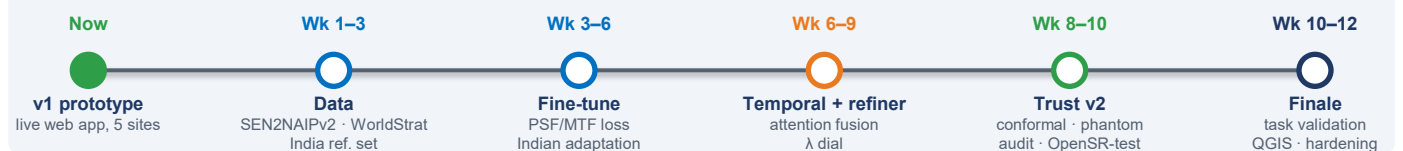
Observed vs inferred detail

multi-date agreement map per pixel

V2

4 · INNOVATION

- **First trust-first SR pipeline for Sentinel-2 in India:** fidelity, detail and uncertainty as separate, auditable layers.
- **Exact ONNX reformulation of the FFT constraint:** a 19 MB model with 9×10^{-7} parity, runs in any browser.
- **Observed vs inferred:** multi-date fusion + conformal intervals turn SR into evidence, not decoration.
- **Sovereign & offline:** no server, no foreign VHR dependency; air-gapped ready.

5 · IMPLEMENTATION ROADMAP

FEASIBILITY, IMPACT & VALIDATION

◆ Potential challenges and risks · Strategies for overcoming these challenges

Risk	Level	Mitigation
Hallucinated detail	HIGH	FFT hard constraint + σ map; flagged, never hidden; phantom audit (v2)
Domain gap (US-trained)	HIGH	Fine-tune on Indian refs (Cartosat via NRSC/NTRO, WorldStrat)
Clouds / monsoon	MED	12-month SCL-scored search; multi-date fusion; SAR guidance (v2)
No 2.5 m ground truth	MED	Self-consistency now; OpenSR-test + task-level tests (v2)
Cross-sensor misalignment	MED	Co-registration + shift-tolerant loss in training
Scale & compute	LOW	Browser for on-demand; tiled GPU batch for state-scale mosaics

◆ Validation metrics · v1 measured on 5 Indian study areas (TTA = 8)

Metric	Mean	Range	Target
Consistency PSNR, 10 m bands	51.1 dB	49.2 – 53.1	> 40 dB
Consistency PSNR, 20 m bands	36.8 dB	34.7 – 40.0	> 33 dB
Spectral angle (SAM), 10 m	0.48°	0.23 – 0.62	< 1.5°
NDVI deviation (MAE)	0.0066	0.003 – 0.009	< 0.02
Edge-energy gain vs bicubic	1.17×	1.14 – 1.21	> 1.1×
Pixels flagged $\sigma > 0.01$	0.05 %	0 – 0.19	< 5 %
Browser vs Python parity	0.02 dB	same scene	< 0.1 dB

v2 reference validation: OpenSR-test and SEN2NAIPv2 cross-sensor pairs (PSNR · SSIM · LPIPS · SAM), India Cartosat set, and building / road / field-boundary / change-detection F1.

◆ Analysis of the feasibility of the idea



Technical

HIGH

- Prototype live; 5 Indian sites validated
- 19 MB model · ≈ 6 s per 1.28 km tile, laptop CPU



Economic

HIGH

- Data free (Copernicus), 5-day revisit
- Hosting ₹0 (static); training on free GPUs



Operational

HIGH

- No install: open a link, or run air-gapped
- Standard GeoTIFF output for GIS workflows



Legal & data

HIGH

- Model CC0-1.0; Sentinel data open licence
- No personal data; nothing leaves the device

◆ Potential impact on the target audience



Defence & intelligence

site-level look without VHR tasking



Disaster management

landslide / flood extent at 2.5 m



Agriculture

field-level NDVI, smallholder bunds



Urban planning

buildings, roads, encroachment



Water & environment

ponds, shorelines, forest loss



Research & education

open, runs on a student laptop

◆ Benefits of the solution (social, economic, environmental, etc.)



Social

- Faster, better-targeted relief
- Field-level farm advisories



Economic

- 2.5 m-class from free data
- No per-km² VHR licences



Environmental

- Finer land & water monitoring
- Early landslide / erosion maps



Strategic

- Sovereign, offline capability
- No foreign data dependency

❖ Details / Links of the reference and research work

Key findings from the literature → our design decisions

SEN2SR [1]: a low-frequency hard constraint gives radiometric consistency; > 15 M parameters brought no significant gain.

→ **Keep the constraint; compact 1.5–15 M backbone.**

GeoSR-Bench [13]: PSNR / SSIM gains often do not track downstream-task gains, and can even run opposite.

→ **Validate on buildings, roads, bunds, change.**

LDSR-S2 [6]: the only prior S2 SR with pixel-wise uncertainty, but diffusion is slow.

→ **TTA σ now; diffusion only as optional refiner.**

SEN2NAIP [9]: 2,851 real + 17,657 synthetic pairs; real pairs are mis-registered.

→ **Synthetic pre-train, shift-tolerant fine-tune.**

HighRes-net, WorldStrat, MuS2 [8, 10, 11]: multiple revisits add true sub-pixel information.

→ **Temporal attention front-end in v2.**

References by domain

Super-resolution methods

- [1] Aybar et al., 2025. SEN2SR, radiometrically & spatially consistent SR for Sentinel-2. Remote Sens. Environ.
- [2] Wan et al., 2024. SPAN: Swift Parameter-free Attention Network. CVPRW.
- [3] Liang et al., 2021. SwinIR. ICCVW.
- [4] Wang et al., 2018. ESRGAN. ECCVW.
- [5] Yue et al., 2023. ResShift, efficient diffusion SR. NeurIPS.
- [6] Donike et al., 2025. Trustworthy S2 SR with latent diffusion.
- [7] DiffFuSR, 2025. All-band S2 SR. arXiv 2506.11764.

Multi-image, data & benchmarks

- [8] Deudon et al., 2020. HighRes-net. arXiv 2002.06460.
- [9] Aybar et al., 2024. SEN2NAIP. Scientific Data.
- [10] Cornéise et al., 2022. WorldStrat. NeurIPS D&B.
- [11] MuS2, 2023. Multi-image S2 SR benchmark. Scientific Data.
- [12] ESA OpenSR. opensr-test benchmark (GitHub).
- [13] GeoSR-Bench, 2026. Downstream-task SR benchmark. arXiv 2605.00310.

Uncertainty & data / tools

- [14] Lakshminarayanan et al., 2017. Deep ensembles. NeurIPS.
- [15] Wang et al., 2019. Test-time augmentation uncertainty. Neurocomputing.
- [16] Angelopoulos & Bates, 2021. Conformal prediction. arXiv 2107.07511.
- [17] ESA. Sentinel-2 MSI user guide (bands, MTF).
- [18] Copernicus Data Space · Element 84 Earth Search.
- [19] ONNX Runtime Web (WASM / WebGPU).

PROJECT RESOURCES



Live web app

uchiadzero.github.io/Deep-Learning-Super-Resolution-Mapping



Source code (GitHub)

github.com/UchiadZero/Deep-Learning-Super-Resolution-Mapping



Technical design document

TECHNICAL_DESIGN.md in the repository



Demo video

[\[add YouTube link\]](#)



Implementation plan

IMPLEMENTATION_PLAN.md in the repository

Model credit: SEN2SR-Lite (ESA OpenSR, CC0-1.0) is the v1 baseline. The pipeline, trust layer, validation, ONNX engine and web app are ours.